MYCAMPUSVIEW ACADEMY · ROBOTICS 101

Arduino cheat sheet

The commands used in this course

Code	What it does	Usually in
`pinMode(pin, OUTPUT)`	Set a pin as output (motors, LEDs) or INPUT (sensors)	setup()
`digitalRead(pin)`	Read a sensor: HIGH or LOW	loop()
`digitalWrite(pin, HIGH)`	Switch a pin on (HIGH) or off (LOW)	loop()
`analogWrite(pin, 0–255)`	Set motor speed with PWM (pins 5 and 6)	loop()
`delay(ms)`	Wait; 1000 ms = 1 second	anywhere
`Serial.begin(9600)`	Start talking to the Serial Monitor	setup()
`Serial.println(x)`	Print a value and a new line	anywhere
`if (…) { } else if (…) { } else { }`	Choose what to do	loop()
`&& || !`	AND, OR, NOT	conditions
`const int NAME = 2;`	Give a pin or setting a name	top of sketch
`// comment`	A note for humans; the robot ignores it	anywhere

Every sketch has two parts

```
void setup() {
  // runs once when the robot switches on
}

void loop() {
  // runs again and again, forever
}
```

Common errors

Error says	Usually means
expected ';' before …	A missing semicolon on the line above
'leftspeed' was not declared	A spelling or capital-letter mismatch
expected '}' at end of input	A missing closing bracket
Port not found / upload failed	Wrong port, or the USB cable is loose